

Quiz 4
CSE471: System Analysis and Design
SET-B

Name:

ID:

Total Marks: _____ / 20

Time: 25 minutes

Data Store Design & Program Design (CO2)

Q1. You should show the proper units for all the questions, detailed calculations, and/or formulas.

Attributes	Average Size (bits)
TAX	16
VAT	104
Reg No.	16
TIN	72
Bill No.	32
Transaction ID	32
SSN	64
Expiry	56

- a. Find the **Record Size** in **bytes**. [2]

$$\begin{aligned}\text{Record Size} &= \text{Summation of all the attributes' average size} \\ &= ((16 + 104 + 16 + 72 + 32 + 32 + 64 + 56) / 8) \text{ bytes} \\ &= 49 \text{ bytes}\end{aligned}$$

- b. The system's overhead will be 0.3. Calculate the **Total Record Size** in **bytes**. [2]

$$\begin{aligned}\text{Total Record Size} &= \text{Record Size} + (\text{Record Size} * \text{overhead}) \\ &= 49 + (49 * 0.3) \text{ bytes} \\ &= 63.7 \text{ bytes}\end{aligned}$$

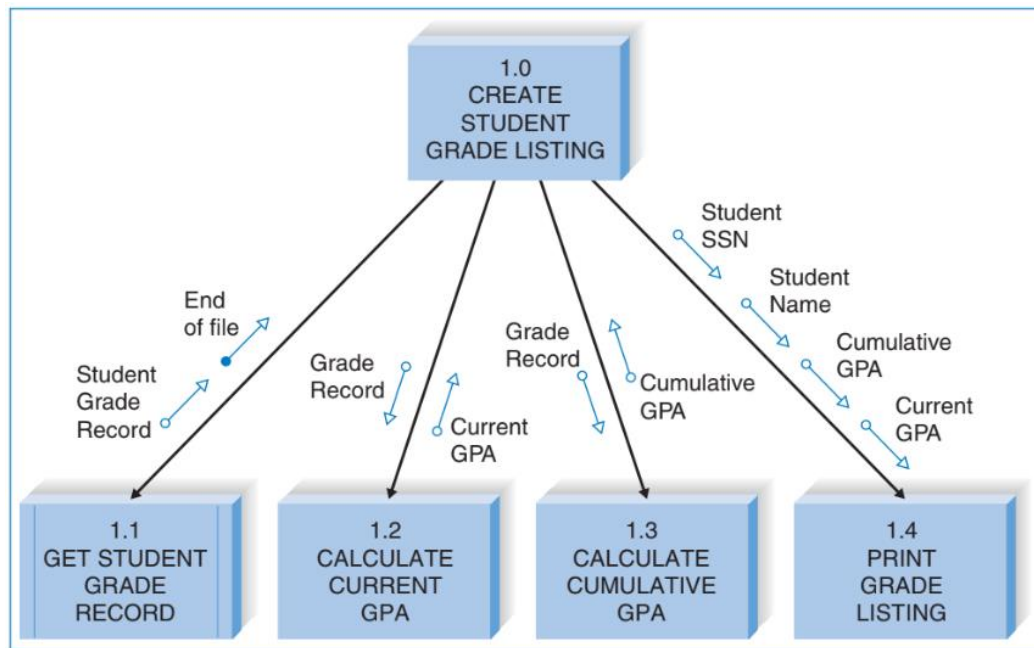
- c. The Initial Table Size is 50,000. Calculate the **Initial Table Volume** in **bytes**. [2]

$$\begin{aligned}\text{Initial Table Volume} &= \text{Initial Table Size} * \text{Total Record Size} \\ &= (50,000 * 63.7) \text{ bytes} \\ &= 3,185,000 \text{ bytes}\end{aligned}$$

- d. Calculate the **Final Table Volume** in **bytes** after 36 months; quarterly growth is 3000.[4]

$$\begin{aligned}\text{Final Table Volume} &= \text{Initial Table Volume} + (\text{Growth per unit} * \text{Total Record Size} * \text{total time}) \\ &= (3,185,000 + (1000 * 63.7 * 36)) \text{ bytes} \\ &= 5,478,200 \text{ bytes}\end{aligned}$$

Q2.



- a. How many distinct data and control couples are there? [2]

* Distinct means different types [Student Grade Record = Grade Record]

Ans: Data – 5, Control - 1

- b. What kind of module is 1.0? [2]

Ans: Control module

- c. Which subordinate modules do you see? [2]

Ans: 1.1(Get Student Grade Record), 1.2 (Calculate Current GPA), 1.3 (Calculate Cumulative GPA), 1.4 (Print Grade Listing)

- d. What are the fan-in and fan-out numbers for 1.1 and 1.4? [2]

Ans: 1.1 – Fan-in: 1, Fan-out: 0, 1.4 – Fan-in: 1, Fan-out: 0

- e. How many modules can call 1.0 at max? [2]

Ans: 1